

Tumor Morphology and Phenotypic Evolution Driven by Selective Pressure from the Microenvironment

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SUMMARY

Emergence of invasive behavior in cancer is life-threatening, yet ill-defined due to its multifactorial nature. We present a multiscale mathematical model of cancer invasion, which considers cellular and microenvironmental factors simultaneously and interactively. Unexpectedly, the model simulations predict that harsh tumor microenvironment conditions (e.g., hypoxia, heterogeneous extracellular matrix) exert a dramatic selective force on the tumor, which grows as an invasive mass with fingering margins, dominated by a few clones with aggressive traits. In contrast, mild microenvironment conditions (e.g., normoxia, homogeneous matrix) allow clones with similar aggressive traits to coexist with less aggressive phenotypes in a heterogeneous tumor mass with smooth, non-invasive margins. Thus, the genetic make-up of a cancer cell may realize its invasive potential through a clonal evolution process driven by definable microenvironmental selective forces. Our mathematical model provides a theoretical/experimental framework to quantitatively characterize this selective pressure for invasion and test ways to eliminate it.

INTRODUCTION

It is widely accepted that tumors evolve by clonal selection of cell populations that proliferate in an unconstrained manner, accumulate mutations, and compete for nutrients or space (Nowell, 1976). In the current paradigm, as a result of mutation in critical genes, the cancer tissue becomes increasingly aggressive, locally invasive, and ultimately metastatic (Fearon and Vogelstein, 1990). While it is easy to envision the elements of selection for nutrients

or space in tumor growth (Hanahan and Weinberg, 2000), selective pressure for invasion and metastasis is less obvious (Liotta and Kohn, 2003). The prevailing view is that geno/phenotypes that meet all requirements for metastasis are a rare occurrence in a primary tumor, in agreement with the observed inefficiency of metastasis (Liotta and Kohn, 2003). However, this view has been seriously challenged by recent data showing that metastatic tumors bear a molecular signature that also dominates in the originating primary tumor (Ramaswamy et al., 2003). Furthermore, there is little understanding of the paradox that certain cancers (e.g., small cell lung cancer) metastasize early, while still small, whereas other tumors (e.g., breast adenocarcinoma) metastasize as a function of size. Thus, the mechanism of selection for invasive cancer cells remains an open fundamental question in cancer biology.

It is generally agreed that cancer invasion is the outcome of a complex, multifactorial interplay between cancer cells and the host tissue microenvironment since explanations of invasion based on single factors appear to be inadequate (Liotta and Kohn, 2001). Mathematical modeling is an ideal approach for teasing apart mechanisms of cancer invasion because it can simultaneously and quantitatively consider interactions between multiple factors (Araujo and McElwain, 2004; Chaplain, 1996). Here we report from a multiscale mathematical model of cancer invasion results that show good predictive ability and intuitive links with cancer pathology and experimentation. In our mathematical model, the behavior of cancer cells is based on a set of equations for how individual cells grow, divide, move (both random and directed motion), mutate, and interact with each other (cell-cell adhesion, crowding) and the microenvironment (including consumption of nutrients and production of enzymes) (for a complete mathematical derivation, see Anderson, 2005). From a mathematical point of view, some of these variables are continuous (describing densities or concentrations rather than individual entities) and some discrete (describing single entities). Therefore, our primary approach is based on a hybrid discrete-continuum (HDC) model in which both discrete stochastic and continuum

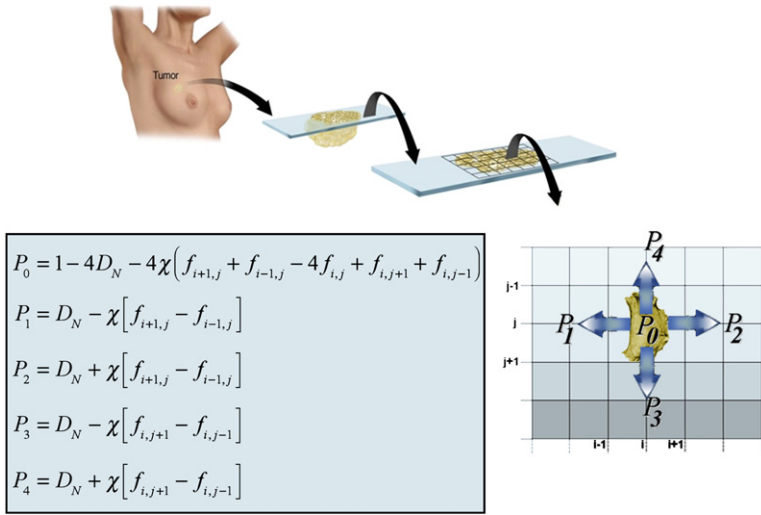


Figure 1. Movement Probabilities for Tumor Cells in the HDC Model

Intuitively, the model can be thought of as representing a slice of tumor on a mathematical lattice. The four equations P_{1-4} represent the probability of a single cell located at the lattice coordinates (i,j) moving to a neighboring point on the lattice. Each cell is assumed to have an intrinsic unbiased motility [of rate D_N] and biased motility driven by haptotaxis (of rate χ) toward higher ECM concentrations (f). The equation P_0 represents the probability of a cell being stationary. It takes into account the situation when a single cell does not experience a gradient between neighboring points because they contain equal concentrations of ECM; if neighboring points contain higher (lower) ECM concentrations, the probability of being stationary is diminished (increased) by the sign and magnitude of the term $(f_{i+1,j}^q + f_{i-1,j}^q - 4f_{i,j}^q + f_{i,j+1}^q + f_{i,j-1}^q)$. For a full mathematical derivation of these equations, please see [Anderson, 2005](#).

deterministic variables are coupled ([Anderson, 2005](#); [Anderson and Chaplain, 1998](#); [Anderson et al., 1997, 2000](#)). Because of its hybrid nature (cells treated as discrete entities and microenvironmental parameters treated as concentrations), the model can be directly linked to experimental measurements of those cellular and microenvironmental parameters recognized by cancer biologists as important in cancer invasion. Furthermore, the fundamental unit of the model is the cell, and the complex collective behavior of the tumor emerges as a consequence of interactions between factors influencing the life cycle and movement of individual cells.

RESULTS

Since we are interested in predicting invasion, our model produces simulations of the growth of a vascularized solid tumor (though, for simplicity, mathematical representation of vascularization is kept to a minimum at this stage). To describe cell motility in mathematical terms, cells are located on a two-dimensional lattice ([Figure 1](#)) that represents the tissue domain in which cells live, including extracellular matrix and other factors. Cells are allowed to move to free neighboring positions on the lattice according to movement probabilities defined by the equations in [Fig-](#)

[ure 1](#). The model calculates movement probabilities by first considering cell-matrix interactions that may drive motility. However, before any movement can occur, the influence of other crucial cellular parameters such as cell-cell adhesion and proliferation is also considered.

To include these parameters, each cell is given an individual life cycle flow chart that takes into account both cell phenotype and microenvironmental influences ([Figure 2](#)). Its components include both basic metabolic processes, such as proliferation and oxygen consumption rates, as well as motility-related ones, such as the propensity to undergo haptotaxis and engage in cell-cell adhesion. In addition, interactions with the microenvironment are governed by three key continuous variables: extracellular matrix (ECM) macromolecule concentration (denoted by f), the concentration of the matrix-degrading enzyme (MDE) (denoted by m) secreted by the cell to break down the ECM, and oxygen concentration (denoted by c). Note that oxygen could be representative of any nutrient or nutrients in general necessary for tumor cell survival. The ECM is the macromolecular environment surrounding cells and so has distinct composition and structure depending on the location of the tumor (e.g., lung, breast, prostate, bone, etc.). The equations describing cell interactions with ECM, MDEs, and oxygen are:

$$\begin{aligned}
 \text{rate of change of MDE} \quad \frac{\partial m}{\partial t} &= \underbrace{D_m \nabla^2 m}_{\text{diffusion of MDE}} + \underbrace{\mu N_{i,j}}_{\text{production of MDE by tumour cell}} - \underbrace{\lambda m}_{\text{decay of MDE}} \\
 \text{rate of change of Matrix Macromolecule (MM)} \quad \frac{\partial f}{\partial t} &= - \underbrace{\delta m f}_{\text{degradation of MM by MDE}} \\
 \text{rate of change of oxygen} \quad \frac{\partial c}{\partial t} &= \underbrace{D_c \nabla^2 c}_{\text{diffusion of oxygen}} - \underbrace{\alpha c}_{\text{decay of oxygen}} - \underbrace{\gamma N_{i,j} c}_{\text{oxygen consumption by tumour cell}} + \underbrace{\beta f}_{\text{production of oxygen by MM}}
 \end{aligned}$$

where D_m and D_c are the MDE and oxygen diffusion coefficients, respectively; μ , λ , δ , α , γ , and β are positive constants. The $N_{i,j}$ term in each of these represents a tumor cell located at lattice position (i,j) and is either one if a cell is present or zero if it is not. Oxygen is assumed to diffuse, decay naturally, and be consumed by the tumor. To minimize the modeling of tumor vascularization, oxygen production is proportional to the ECM macromolecule (MM) density. However, it must be stressed that uncoupling oxygen production from ECM density produces similar results in the simulations (data not shown). The multiscale nature of the HDC model facilitates incorporation, at a later time, of a complex angiogenesis supply model (Anderson and Chaplain, 1998), more sophisticated models for cell movement, more detailed modeling of intracellular processes, or other variables.

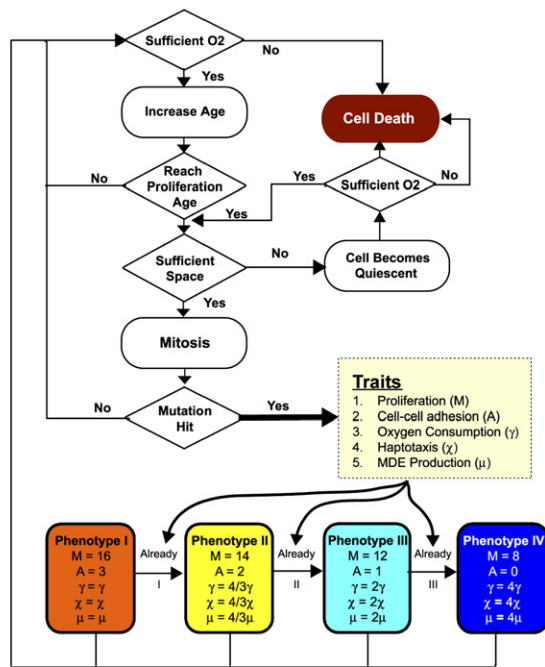
The life-cycle flow chart is the core engine of the HDC model and provides a natural link between mathematics and biological experimentation (Figure 2). This is because each cell behaves on its own, based on a phenotype determined both by genetic make-up and microenvironmental interactions. The ability to assign a phenotype to each individual cell is a fundamental property of our HDC approach. The advantage of the HDC approach with phenotypes is perhaps best illustrated by its evolution from a previous, purely continuous treatment (Anderson et al., 2000) in which all variables, including cells, were treated mathematically as continuous and solved as partial differential equations (changing in both space and time). In this earlier version (Figure 3, upper panels), simulation of tumor cells growing in a “bumpy” ECM (see below and legends to Figures 3 and 4A) mimics the contours of the matrix itself. In this model cells are a continuous variable and are represented as an average. Therefore, individual cell processes crucial to cancer, e.g., cell-cell adhesion, cannot be captured, thus limiting the link to experimental observation or validation. In a more evolved version, discretizing cells (Figure 3, middle panels) allows for better experimental validation of the model and allows for independent behavior of individual cells, e.g., competition for space. However, the model still lacks realism because all cells are assumed to be identical, in spite of the well-known coexistence of heterogeneous cell populations within tumors. For example, in the model run as in Figure 3 (middle panels), cells are all considered as having no cell-cell adhesion and the same proliferation rate. In the present version, the advantages of discretizing cells become dramatically apparent when individual phenotypes are assigned to each of the cells (Figure 3, lower panels). The phenotypes are defined by mutating traits (Figure 2 and Table 1A) based on current views of the invasive phenotype (Hanahan and Weinberg, 2000). In the life-cycle flow chart, as in reality, mutations are propagated at the time of cell division (Figure 2). The simulation with individual cell phenotypes produces a tumor fingering pattern that is reminiscent of invading cancer (Figure 3, lower panels). The fingering is a direct outcome of the way individual cells interact with ECM and with each other in terms

of adhesion. Cell death is also simulated as a consequence of oxygen deprivation, resulting in a tumor core of dead cells. It should be emphasized that individual cell traits can be included or excluded from the life cycle, based on their relative importance in the invasion process, as determined by experimentation.

To apply the model and simulate cancer invasion, initially we adopted a linear mutation scheme (Figure 2A). In this scheme, cells mutate unidirectionally along a linear pathway of four increasingly aggressive phenotypes with predefined traits and values shown in Table 1A. Such a scheme is reminiscent of the current models of tumor progression, generally referred to as “vogelgrams” (Fearon and Vogelstein, 1990). The most aggressive phenotype (type IV) is characterized by shortest lag time to proliferation, highest oxygen consumption, highest ECM-degrading enzyme production, highest haptotactic migration rate, and no cell-cell adhesion (Table 1A). We simulated tumor cell proliferation in three distinct microenvironments, characterized by initial ECM conditions referred to as homogeneous, “bumpy,” and “grainy” (Figure 4A, upper panels). Bumpy and grainy ECM represent density variations at either the macroscale or at the same scale as the cells, respectively. Figure 4A, middle panels, shows the tumor morphology after simulated growth with the linear mutation scheme for 200 generations, approximately 4.5 months. Two distinct tumor morphologies emerge in the simulations, depending on the ECM structure: a mass with smooth margins in the homogeneous matrix, as opposed to a fingering mass in the heterogeneous matrices. There is an obvious resemblance between the tumor morphology in heterogeneous ECM simulations and pathological specimens of locally invasive carcinoma. However, even though the tumor morphology is distinct, segregation of phenotypes in the three simulations is the same: dead cells in the middle, and the most aggressive phenotype on the boundary. This is not entirely surprising since dominance of the most aggressive phenotype in the linear mutation scheme is essentially predetermined owing to its unidirectionality. Therefore, as the simulation proceeds in time, as soon as cells mutate into phenotype IV, the most aggressive, they tend to dominate the tumor mass. This is somewhat realistic: as mentioned above, the progressive move of tumors toward increasingly aggressive phenotypes is currently explained by “vogelgram” models of accumulation of mutations (Fearon and Vogelstein, 1990; Michor et al., 2004). On the other hand, it is also a common observation that tumors are comprised of genetically heterogeneous populations of cells (Fidler, 2003), which may undergo complex competing interactions during tumor evolution.

In order to eliminate the phenotype selection bias of the linear mutation scheme, and to introduce a larger element of phenotype diversity, we then adopted a random mutation scheme (Figure 2B) in which cells are allowed to “jump” without restrictions to one of 100 predefined phenotypes. These phenotypes are generated by randomly picking trait values with the same upper and lower bounds

A LINEAR MUTATION SCHEME



B RANDOM MUTATION SCHEME

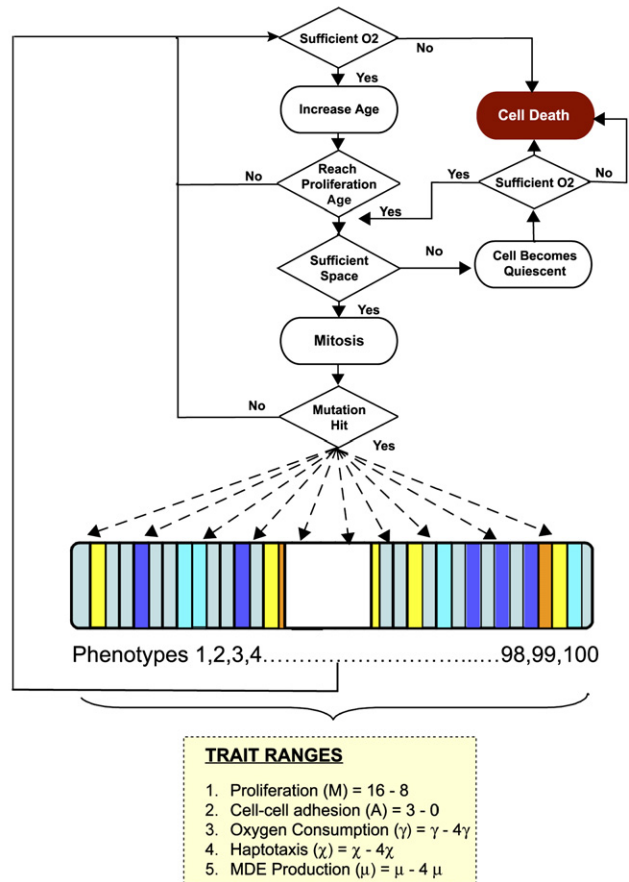


Figure 2. Cancer Cell Life Cycle and Mutation Schemes

In the HDC model, the cancer cell life cycle enables decisions to be made about key functions, e.g., life, death, mutations, based on the interaction between individual cell phenotypes and the local microenvironment. The first step in the cycle is examination of the local microenvironment for each cell, e.g., oxygen concentration and proximity to other cells. Then, dependent on the individual cell phenotype, cells proliferate, die, or become quiescent. Mutations to another phenotype have a small probability of occurring at the time of cell division. Mutations are represented according to two distinct strategies: (A) Linear mutation scheme. In this scheme, cell phenotypes evolve via mutation in a linear direction along four increasingly aggressive, predetermined phenotypes (types I–IV), defined by traits indicated in the boxes. Once cells have reached type IV, they cannot mutate further or revert to a less aggressive phenotype; (B) Random mutation scheme. In this scheme, cell phenotypes are randomly chosen at the time of mutation from one of 100 predetermined phenotypes (initiated at the beginning of each simulation). At each mutation event, a cell can jump to any of the 100 phenotypes, without regard to the parent phenotype. To some degree, this scheme constitutes a simplified way of representing genetic instability.

as the linear mutation scheme (Table 1A). In simulations for the same timeframe (Figure 4A, lower panels, and see Movies S1–S3 available with this article online), the morphology of tumors growing in the random mutation regime is similar to that of the linear mutation results (Figure 4A, middle panels): smooth margins on homogeneous matrix and fingering on both types of heterogeneous matrices. In contrast to the linear mutation, however, the random mutation scheme allows determination of which cell phenotypes are dominant at various points in the simulation without preselection bias.

A striking result of the random mutation scheme is that even though each of the 100 phenotypes is equally likely to be chosen (via a mutation), only a select few actually dominate (Figure 4B; Table 1B). In general, at the end of

the simulations, phenotypes selected by random mutation have the following traits: (1) zero cell-cell adhesion; (2) relatively low oxygen consumption; (3) high proliferation rate; and (4) high haptotaxis coefficient (Table 1A). These characteristics are very similar to the most aggressive phenotype in the linear mutation scheme, with the exception of oxygen consumption (Table 1A).

These results indicate that either scheme provides a realistic outcome, i.e., evolution of tumors toward increasingly aggressive phenotypes. In both cases, it is important to realize that the heterogeneous conditions (bumpy or grainy matrix) drive selection toward a few dominant, very aggressive clones. In contrast, the homogeneous environment with the random mutation scheme allows for coexistence of many phenotypes, more or less aggressive.

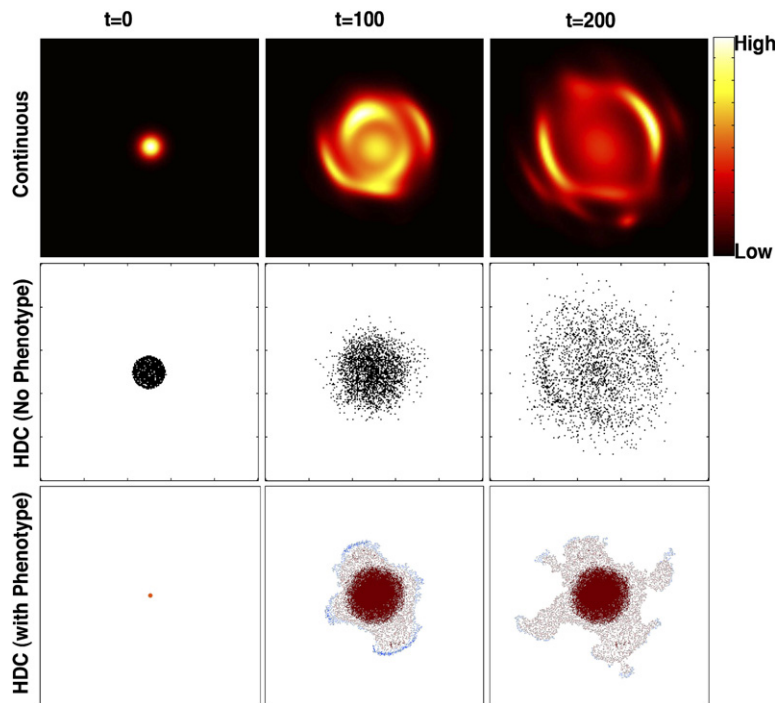


Figure 3. Hybrid Model Development

Each row shows the simulation results from three different stages of the HDC model development. In these simulations, the spatiotemporal evolution of cells growing and invading on a bumpy ECM matrix (matrix composition shown in Figure 4A, middle panel top row) is driven by the following models: row 1: purely continuous partial differential equations model where cells and other variables are considered as densities or concentrations (colors represent cell density, as defined by the heat map on the right); row 2: hybrid model where cells are considered as discrete (although without a phenotype) and the other variables as continuous concentrations; row 3: hybrid model in which cells have a phenotype and can mutate to different phenotypes (red = dead cells; blue = the most aggressive type IV phenotype, which comes to dominate the population over time).

This difference appears to correlate with an emergent property of the corresponding tumor mass: under heterogeneous conditions, tumor morphology is invasive (fingering margins), whereas under homogeneous conditions morphology is noninvasive (smooth margins).

This correlation is intriguing, but fingering morphology in the heterogeneous ECM could merely be due to superior ability to exploit local terrain by cells with higher haptotaxis coefficients. On the other hand, it could be that heterogeneous ECM is a “harsh” environment that acts as a selective force for better-adapted phenotypes. To test this possibility, we considered a different harsh environment, low oxygen (Figure 5A). An advantage of oxygen is that its value can be manipulated in the course of the simulation (this may be also possible from an experimental perspective). To avoid killing the tumor population completely we grew *in silico* tumors in an oxygen-rich environment (until $t = 40$). We then dropped the oxygen concentration to 25% of what it was and almost immediately tumor fingers begin to emanate from the smooth circular core. Afterwards, oxygen was treated as follows: $t = 201$ – 240 (high); $t = 241$ – 350 (low). The outcome is striking: under harsh conditions, when oxygen is in short supply, the morphology of the tumor is invasive ($t = 200$; $t = 350$). In mild conditions, the morphology is noninvasive ($t = 40$; $t = 240$). Note that in these simulations the initial ECM distribution was homogenous.

An important benefit of this simulation approach is that at the same time as we simulate tumor growth we also track the numbers of the different phenotypes in the population. Figure 5B shows the resulting phenotype distribution for this simulation. The difference between the

numbers of phenotypes in the high and low oxygen micro-environments is obvious and clearly shows the selective ability of the harsh microenvironment. It is noteworthy that the same three aggressive phenotypes dominate during both periods of low oxygen conditions and have similar aggressive traits as those selected by heterogeneous ECM (Table 1A). In contrast, under mild conditions (high oxygen) many clones coexist, marked by different levels of aggressiveness, similar to the homogeneous ECM. These results strengthen the notion that microenvironment conditions decide the invasive outcome of a tumor and suggest a mechanism based on selection of a few dominant phenotypes out of a randomly generated mixture. Invasion appears to be an emergent property of this selection process.

Cell-cell adhesion based on homotypic interactions is of special importance in tumor progression because of contact inhibition and because of the association of E-cadherin loss with metastasis and the upregulation of other homotypic cell-cell adhesion molecules. Our simulations also point to the importance of cell-cell adhesion since the phenotypes selected by the harsh microenvironment always have zero cell-cell adhesion. We therefore modified the HDC code in order to simulate more complex cell-cell adhesion interactions. The overall outcome is that even relatively minor additions to the complexity of cell-cell adhesion phenotype traits substantially affects both tumor evolution and morphology. For example, we introduced the possibility that tumor cell clones may express either of two distinct homotypic adhesion pairs (by analogy to the E- to N-cadherin switch that occurs in tumor progression; Hazan et al., 2004). These simulations were

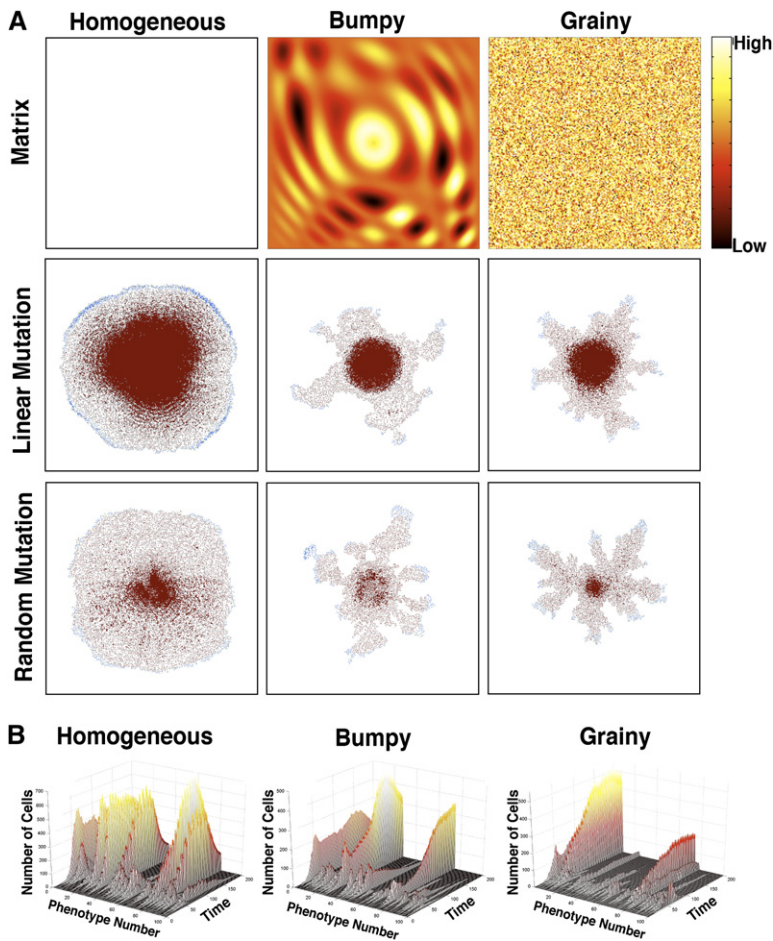


Figure 4. Effect of Extracellular Matrix Structure on Tumor Morphology and Phenotype Selection

(A) Heterogeneous ECM induces invasive tumor morphology. Simulations were run with the current HDC model with three different initial ECM distributions (upper row), using either the linear (middle row) or random (lower row) mutation schemes. Density of ECM is represented by coloration, as depicted in the color bar on the right, and is termed homogeneous, “bumpy,” and “grainy” as indicated. Cell coloration reflects dead cells (red) or cell-cell adhesion (zero cell-cell adhesion = blue). For the linear mutation scheme, blue also represents the most aggressive phenotype (type IV, see Table 1A for representative dominant phenotypes). For the random mutagenesis scheme, see the entire simulations in Movies S1–S3.

(B) Heterogeneous ECM selects few, aggressive tumor cell phenotypes. Phenotype distributions for the random mutation scheme simulations shown in Figure 4A, row 3 are depicted here. The number of viable cells for each particular phenotype in the population over time is represented in a three-dimensional plot for each ECM condition. The coloration is a “heat map” that represents cell number. Fewer phenotypes (1–3, compared with 5–7 in the homogeneous matrix) survive in the heterogeneous ECM conditions, with the fewest (1–2) in the grainy ECM. ($n = 20$ simulations for each condition, representative results shown).

run under the random mutation regime, so that each of the 100 initial phenotypes is assigned an adhesion type (A or B), with a corresponding adhesion value (0–3). Cells of adhesion type A can only adhere to A, and B to B.

In comparison to simulations with a single adhesion type (Figure 5), we observed the following (Figure 6): (1) Tumors with mixed adhesion types are smaller and more compact (Figures 6A and 6B); (2) At the first round of selection with hypoxia, cells with less “aggressive” traits (higher cell-cell adhesion) still coexist (Figures 6A and 6C); (3) After the second round of selection, the mixed adhesion phenotypes have segregated into invasive fingers, or lobes, of either one type or the other (Figure 6B). At this late stage, tumor morphology is similar to the first round of hypoxia selection in the uniform adhesion scheme (Figure 5A). These results show that the HDC model (1) can substantially capture biological complexity, e.g., as represented by mixed cell-cell adhesion molecules; (2) is multiscale, i.e., molecular level events are translated into effects at the tissue scale; (3) is suitable for validation by current technology, whose preferred approach is based on single gene changes, e.g., by manipulating expression of specific cell-cell adhesion genes to generate cell populations containing mixed homotypic adhesion pairs and

determine their impact on tumor morphology and phenotype selection.

DISCUSSION

Implications of HDC Model Predictions

The HDC model makes rather specific predictions on the mechanism of emergence of cancer invasion under selective pressure by the microenvironment. Harsh conditions produce tumors that are morphologically invasive (fingering margins) and are dominated by one (or few) aggressive phenotypes. In the simulations, these two conditions are inextricably linked. In contrast, mild conditions lead to coexistence of aggressive with less aggressive phenotypes, no clonal dominance, and tumors with smooth, noninvasive margins. Again, these two conditions are strictly linked to each other. These predictions suggest that aggressiveness of diverse cancers may be a consequence of tumor growth in the context of a harsh or mild microenvironment. This concept is consistent with and may help explain experimental and clinical findings showing that hypoxia or nutrient deprivation leads to invasive, aggressive tumor behavior (Casanovas et al., 2005; Erler et al., 2006; Frieboes et al., 2006; Kerbel, 1990;

Table 1. Representative Values of Traits for Tumor Cell Phenotypes Selected by the Microenvironments and Summary of Simulation Outcomes

(A)						
Traits	Range	Linear (IV)	Random (Homogeneous)	Random (Bumpy)	Random (Grainy)	Random (Low Oxygen Switch)
Proliferation	16–8 hr	8 hr	9.4 hr	9.6 hr	9.1 hr	8.1 hr
O ₂ Consumption	γ –4 γ	4 γ	2.5 γ	2.1 γ	2.2 γ	2.0 γ
MDE Production	μ –4 μ	4 μ	3.5 μ	2.7 μ	2.9 μ	2.7 μ
Cell-Cell Adhesion	3–0	0	0	0	0	0
Haptotaxis	χ –4 χ	4 χ	3.5 χ	2.7 χ	2.9 χ	2.7 χ
(B)						
Scale	Microenvironment					
	Mild			Harsh		
Molecular/subcellular	Aggressive traits selected			Aggressive traits selected		
Cellular	Many phenotypes selected and coexisting			1–3 phenotypes selected and dominating		
Tumor	No invasive morphology			Invasive morphology		

Pennacchietti et al., 2003). Taking it one step further, the HDC model predicts that invasive tumor properties are reversible under appropriate microenvironment conditions (Figures 5A–5B) and suggests that differentiating therapy aimed at cancer-microenvironment interactions may be more useful than making the microenvironment harsher (e.g., by chemotherapy or antiangiogenic therapy).

HDC Model Variables and Parameters in Relation to Cancer Invasion Biology

In its current version, the HDC model adopts a minimal modeling approach, with four model variables (coupled equations), cells, ECM, MDE, and O₂. Nonetheless, it provides realistic simulations of cancer invasion. Furthermore, overly complex modeling schemes with many coupled equations and parameters make it difficult to extract

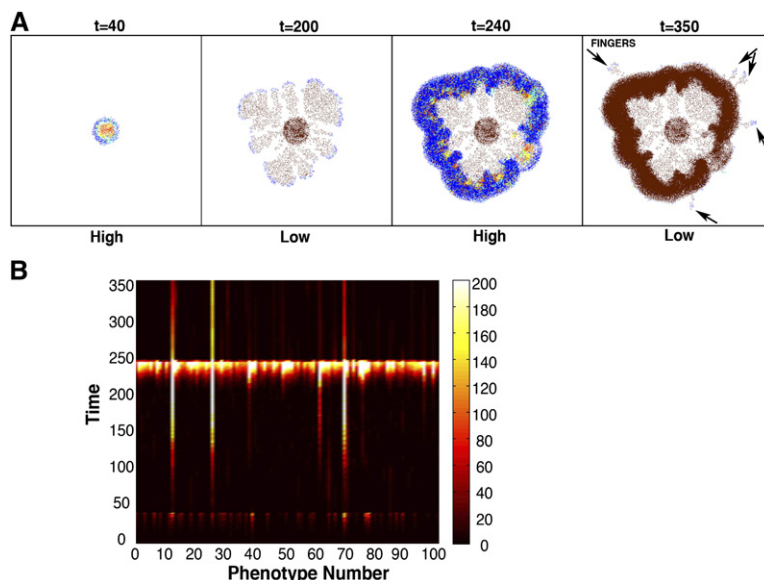


Figure 5. Manipulation of Oxygen Concentrations Reveals a Global Effect of the Microenvironment on Tumor Morphology and Phenotype Diversity

(A) Effect of oxygen concentration on tumor morphology. Simulations were run with the HDC model using a homogeneous ECM distribution and a switching combination of oxygen concentrations. Oxygen is either held constant at 1 throughout the domain (High) or set to 0.25 throughout the domain and allowed to be consumed (Low). From left to right, the tumor morphology in the High (from $t = 0$ –40), Low (from $t = 40$ –200), High (from $t = 200$ –240), and Low (from $t = 240$ –350) are represented. Cell coloration reflects either dead cells (brown) or the cell-cell adhesion value (0 = blue, 1 = cyan, 2 = yellow, 3 = orange). Arrows demarcate the growth of tumor “fingers” in the final low oxygen environment (far right panel). (See entire simulation in Movie S4).

(B) A mild environment (e.g., high oxygen) allows all phenotypes to coexist where a harsh environment (e.g., low oxygen) favors dominance of fewer phenotypes in the random mutation scheme. Phenotype distribution (x axis) of the cells at various time points (y axis) is depicted here. Coloration is a “heat map” and represents cell number, as indicated by the color bar on the right y axis. Note that during high oxygen conditions ($t = 0$ –40 and $t = 200$ –240), all 100 phenotypes are represented whereas during low oxygen conditions ($t = 40$ –200 and $t = 240$ –350), there is a large drop in the diversity of phenotypes that are expressed and the same three phenotypes become dominant again.

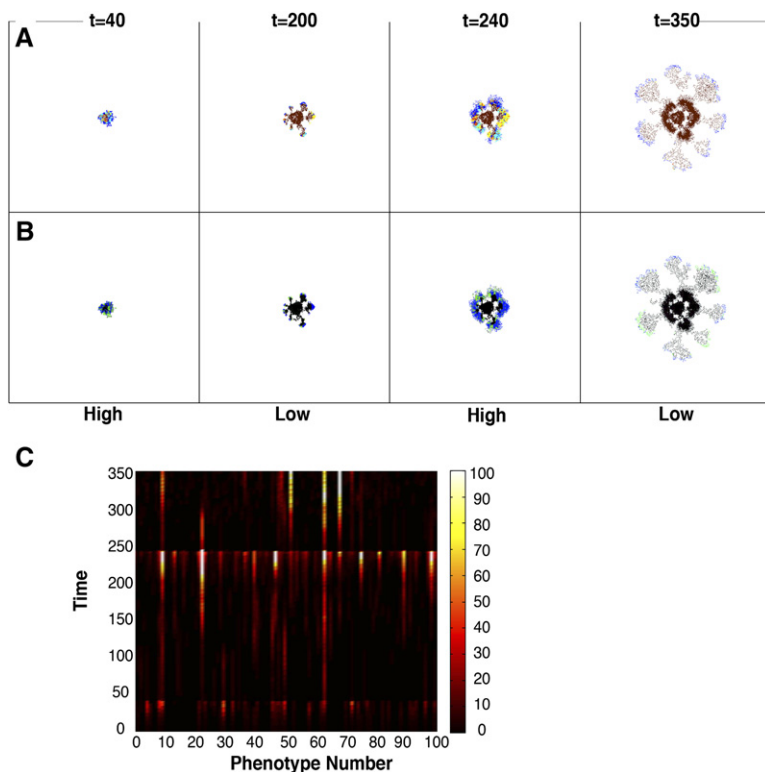


Figure 6. Effects of Homotypic Cell-Cell Adhesion Complexity on Tumor Morphology and Phenotype Diversity

(A) Effect on tumor morphology and size. Simulations were run with the HDC model using a homogeneous ECM distribution and a switching combination of oxygen concentrations. Oxygen is either held constant at 1 throughout the domain (High) or set to 0.25 throughout the domain and allowed to be consumed (Low). From left to right, the tumor morphology in the High (from $t = 0-40$), Low (from $t = 40-200$), High (from $t = 200-240$), and Low (from $t = 240-350$) are represented. Cell coloration reflects either dead cells (brown) or the cell-cell adhesion value (0 = blue, 1 = cyan, 2 = yellow, 3 = orange). (See entire simulation in Movie S5).

(B) Distribution of cell adhesion homotypes. The same simulation as in (A) but now coloration reflects the cell-cell adhesion homotypes (A = blue, B = green), as well as dead cells (black). Note that there is a mix of adhesion homotypes in most parts of the tumor ($t = 40, 200, 240$) which later ($t = 350$) self-segregates into lobes that contain either one or the other homotype.

(C) Effect of homotypic adhesion complexity on phenotype selection under high-low oxygen conditions. Phenotype distribution, represented as in Figure 5B, shows that during high oxygen conditions ($t = 0-40$ and $t = 200-240$),

all 100 phenotypes are represented whereas during low oxygen conditions ($t = 40-200$ and $t = 240-350$), there is a large drop in the diversity of phenotypes. However, unlike Figure 5B, the addition of homotypic adhesion complexity appears to cause some new phenotypes to emerge as dominant at $t = 240-350$.

key individual variables and/or parameters. It also makes it much more difficult to perform experimental validation since each variable has multiple associated parameters to be measured. However, as we experimentally test the model, increasing levels of complexity can be easily introduced because the model is open to incorporation of other variables and experimental data from multiple scales, e.g., the macroscale (tissue), microscale (cells), subcellular, and molecular scales.

The HDC model has certain built-in assumptions and dependent variables. For example, haptotaxis is a key determinant of tumor cell migration and is affected by both the initial ECM concentration and the gradients created by MDE activity. Thus, ECM heterogeneity can dominate tumor morphology and constitute a “harsh” microenvironment. However, in other simulations, low oxygen concentrations in homogeneous ECM also provide a clonal selective force on growing tumors that results in a fingering morphology. The specific microenvironmental parameters and range of concentration values of those parameters that act as selective forces in real life can only be determined by experimentation.

In simulations, a major qualitative change is visualized when gross alterations of the microenvironment are introduced. For instance, homogeneous versus inhomogeneous ECM produces a change from smooth to fingering margins. However, more subtle alterations, e.g., from

bumpy to grainy ECM, produce relatively minor changes in the fingering patterns. This prediction can only be validated by parameterizing the model, i.e., introducing into the model actual measured values for ECM supramolecular organization, oxygen concentration in the tumor tissue, etc. For instance, an approximation for oxygen concentration can be obtained by staining tissue sections with pimonidazole, and areas of hypoxia can be marked by carbonic anhydrase staining. ECM organization could be determined by quantitative staining for distinct ECM molecules in serial histological sections.

To model tumor plasticity in different microenvironments, we resorted to a random mutation scheme (Figure 2B), with 100 random phenotypes. Thus, simulated tumors generate a diversity of cell phenotypes, upon which microenvironment forces operate selection. In this sense, tumors modeled by the random scheme are plastic within the range of the 100 phenotypes, though only the adapted phenotypes survive. For example, simulations under harsh conditions show that tumor heterogeneity is eventually lost (though it can be regained when conditions are mild again).

In the model (Figure 5B), alternating harsh with mild microenvironment conditions results in reversible switching between few or many tumor cell phenotypes, respectively. How could this be achieved biologically? One possibility is that tumor cells, because of intrinsic plasticity,

regenerate multiple phenotypes from a few, based perhaps on metabolic or epigenetic adaptation to the microenvironment. Another possibility is that a totipotent pool of tumor stem cells radiates into many phenotypes subsequently selected according to microenvironment conditions while asymmetrically dividing to maintain the stem pool independent of microenvironment. These, or other possibilities, will have to be tested experimentally, and the experimental outcome can be easily accommodated in the HDC model. For instance, if existence and phenotypic traits of tumor stem cells are better defined experimentally, then the life-cycle flow chart may have to be modified to include a loop for asymmetrically dividing cells with stem quality.

Experimental Validation of the HDC Model

An essential attribute of a useful mathematical model is that its predictions should be testable experimentally, preferably with available or accessible techniques. Key HDC model concepts are that definable microenvironmental forces determine both (1) the shape of a growing tumor (smooth versus fingering margins) and (2) the phenotypic composition of the tumor cell population (Tables 1A and 1B). Both of these concepts can be tested experimentally, using either *in vitro* or *in vivo* methods.

For *in vitro* testing, a defined tumor cell line (e.g., parameterized with respect to rates of proliferation, apoptosis, cell adhesion, etc.) can be expanded from single or few cells under 2D or 3D (semi-solid media) tissue culture conditions that can be varied with respect to chemical or physical microenvironmental variables. For instance, concentration of specific nutrients (glucose, amino acids, or oxygen) or the pH of the culture could be changed. These conditions are easily controlled with current bioreactor technology. More challenging may be to precisely control ECM composition and gradients. However, we have made significant strides toward producing microfluidic devices in which gradients of ECM are easily produced (W. Georgescu, personal communication). The advantage of *in vitro* studies is that the growing tumor cells can be observed or otherwise monitored in real time. Monocellular suspensions can also be easily prepared from these preparations, and phenotype distribution determined, e.g., by flow cytometry or cloning. The disadvantage is that tumor growth *in vitro* is time limited, and one only tests known variables.

In vivo testing of HDC predictions may be the most physiologically relevant form of experimentation due to the complex nature of tumors. The goal is to determine the shape of the growing tumor (infiltrating versus smooth margins) as well as clonality, as a function of microenvironmental selective forces. Certain body compartments may be more suitable to specific microenvironment manipulations: bone, brain, kidney capsule are “bounded” locations into which one could manipulate and monitor conditions of blood flow, inflammation, reactive stroma, or hypoxia. Alternatively, test animals can be subjected to general treatment (e.g., Avastin or other treatment to in-

duce hypoxia), and the local conditions in the tumor can be defined *a posteriori*, e.g., by noninvasive imaging techniques, or histochemistry. This kind of retrospective analysis could also be carried out in human patients, provided a suitable tissue collection is prepared with adequate prognostic and historical information.

Perhaps the most challenging aspect of *in vivo* studies is to define the phenotypic composition and clonality of the growing tumor. This is particularly true of solid tumors (the vast majority of carcinomas, of course) since disaggregating solid tumor specimens to produce monocellular suspensions is cumbersome and may by itself lead to artifacts. There are several possibilities to address this problem. Phenotyping with molecular markers, e.g., for cell-cell adhesion, proliferation, and motility, could be done by conventional immunohistochemical methods on tissue sections, and possibly 3D reconstruction of a tumor specimen, in order to connect phenotypes and their relative dominance with tumor morphology. Clonality can be addressed using mixtures of pre-tagged tumor cells that can be identified in the tumor by histochemical staining or molecular genetic methods.

It is important to note that portions of these experiments have been performed by other investigators but not been organized in a cohesive framework. For instance, recent experiments have addressed the role of hypoxia in promoting aggressive behavior and invasive morphology (Casanovas et al., 2005; Erler et al., 2006; Pennacchietti et al., 2003) but did not examine the clonal or phenotypic composition of resulting tumors. Kerbel and colleagues performed a number of studies examining clonal evolution of primary tumors (Korczak et al., 1988; Rak and Kerbel, 1993) but did not examine invasive behavior at the tissue scale or specific phenotypes selected for. Nonetheless, these experiments underscore the feasibility of HDC model validation.

HDC Model Concepts in Context of Tumor Progression Theories

The concept of tumor microenvironment as a factor in cancer progression has important precedents. Thus, the microenvironment has been proposed to play a part in the promotion phase of classic two-stage tumorigenesis systems (Rubin, 2003) and is perhaps relevant in Paget's seed-soil hypothesis to explain site preference of metastatic growth (Fidler, 2003). Recently, several groups demonstrated the involvement of microenvironmental factors in tumor progression and proposed that the microenvironment may promote (Bhowmick et al., 2004; Kuperwasser et al., 2004) or suppress (Bissell et al., 2002; Matrisian et al., 2001) cancer progression through signaling mechanisms. In contrast, the HDC model indicates that the microenvironment may function as a Darwinian selective agent that favors the emergence of aggressive, invasive clones: simulations under harsh conditions select inevitably for the most aggressive clones, whereas under mild conditions clones with different levels of aggressiveness can coexist. This is reminiscent of evolutionary

selection in theories of quasi-species (Eigen, 1993) or genomic instability (Cahill et al., 1999; Michor et al., 2005).

Nowell proposed in 1976 that the clonal evolution of cancer suggests positive selection of variant sublines, particularly with regard to mutations that confer advantages with respect to competition for space and nutrients or to chemotherapy resistance (Nowell, 1976). Consistent with Nowell's theory, in the HDC model, certain genetic traits (e.g., unrestrained proliferation, motility) are a necessary condition for cancer progression. Importantly, in addition, the HDC model makes the novel prediction that both tumor morphology and the dominance of the invasive phenotypes are a direct outcome of selection by the microenvironment (Figures 4–6). More recently, Weinberg and Hanahan also recognized that cancer progression has the properties of classic Darwinian evolution (Hanahan and Weinberg, 2000), and natural selection in the progression of neoplasia has been advocated in several human systems (Maley and Reid, 2005). The HDC model provides a powerful tool to test these propositions and provide mechanistic insights through quantitative simulations that determine how combinations of microenvironmental and cellular parameters cause invasive tumor growth.

In retrospect, the HDC model appears to directly support the “clonal dominance” theory of cancer metastasis, articulated by Kerbel (Kerbel, 1990) and others in the 1980s and early 1990s. Briefly, Kerbel argued that a major trait of invasive cells is their independence from growth factors, reproducibly arising under harsh microenvironment conditions. Furthermore, he suggested that metastasis is the consequence of such clones taking over and dominating the primary tumor. This view of clonal dominance by the invasive/metastatic cancer cell agrees well with predictions of the HDC model. However, it is in apparent contrast to the prevailing “dogma” that metastatic cells are rare variants randomly occurring in a primary tumor overwhelmingly comprised of metastasis-incompetent cells. These latter views are perhaps best represented by the Fidler metaphor of the metastatic cell as a decathlon winner (Fidler, 2003). In contrast, the HDC model clearly predicts that conditions of harsh microenvironment lead to primary tumors almost entirely comprised of the most aggressive clonal populations. Furthermore, the HDC model predicts that these aggressive populations realize their invasive potential only if harsh microenvironment conditions persist or arise.

Experimental support for the clonal dominance theory comes from studies in which retrovirus-tagged tumor cells were mixed and allowed to evolve in *in vitro* and *in vivo* environments (Korczak et al., 1988). Remarkably, if *in vivo* or *in vitro* conditions were restrictive, metastatic cells always dominated over nonmetastatic tumor cell populations, despite equal growth rates both *in vitro* and *in vivo* (Rak and Kerbel, 1993). Dominance of the metastatic cells within the primary tumor correlated with the development of metastases. The role of a harsh microenvironment in promoting clonal dominance was further tested by Yu et al. in a study that showed that hypoxia promotes dom-

inance of *p53* null cells in mixed tumors of *p53* null and wild-type cells (Yu et al., 2002). These results are consistent with recent clinical studies showing that metastases have the same genotype as primary tumors (Ramaswamy et al., 2003).

The debate among alternative metastasis theories (clonal dominance, dynamic heterogeneity, etc.), fueled by extensive and creative experimental work (Kerbel, 1990; Weiss, 1990), is still unresolved due to the difficulty of reaching clear-cut experimental conclusions from experimental work. This outcome underscores how reductionist approaches, classically used in experimental biology, are inadequate when used alone to capture the complexity of tumor progression and the rise of metastasis. In contrast, we propose that a mathematical simulation approach, such as described here, provides a platform for considering many variables at once and can continue that debate interactively with experimentation.

Future Directions

Above, we provided only a few of many possible validation scenarios, but essentially most of current cancer biology can be appropriately represented and used for validation of the HDC model. Eventually, mathematical representation of molecular networks will be introduced. At the moment, this is impractical for at least three reasons: (1) There is no good agreement as to how even the best known molecular circuitry (e.g., cell cycle) should be represented; (2) Computational power is still limiting to run these sorts of models feasibly in reasonable time; and (3) The effect of a single molecular change on any or all of the model variables is difficult to predict and needs to be defined experimentally (this is in fact a subject of current experimentation in our Center).

A long-term goal of the modeling effort is to be able to predict invasive behavior of individual tumors based on actual measurements of cellular and microenvironmental parameters. If successful, such a predictive tool could be useful in drug discovery, or in the management of cancer patients. For example, high-resolution imaging techniques that quantitate critical properties of cancer cells and of the tumor microenvironment, in combination with mathematical modeling, could be used as an adjunctive aid to customize therapy in individual patients.

Supplemental Data

Supplemental Data include five movies and can be found with this article online at <http://www.cell.com/cgi/content/full/127/5/905/DC1/>.

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